Project #5 – NHTS Midwest Analysis Report

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Introduction

Understanding travel behavior is essential for effective transportation planning. As travel evolves due to changes in technology, income, and lifestyle, agencies must rely on date-driven insight to make informed decisions about infrastructure, sustainability, and mobility. This project analyzes travel behavior using data form the National Household Travel Survey (NHTS 2022). This data set included information on household demographics, vehicle characteristics travel activity, and more. The analysis was conducted with a specific focus on the Midwestern region. The primary objectives of this study were to examine how socioeconomic factors relate to vehicle ownership patterns and travel behavior. By integrating multiple levels of the NHTS dataset such as household income, vehicle, and travel variables this analysis also explores differences in fuel type adoption, vehicle composition, and travel activity across income groups. The findings of this research aim to provide insight into how income influence transportation choices which can support future panning efforts related to alternative fuel adoption, and infrastructure development.

Research Questions

This study is guided by two primary research questions that observe the relationship between household income, vehicle characteristics, and travel behavior within the Midwestern region.

Research Question 1

How does the adoption of alternative fuel vehicles (AFV) vary across household income groups in the Midwest?

The transition towards AFV such as hybrid and fully electric powertrains is an important component of sustainable transportation planning. However, adoption of these technologies may be influenced but factors such as household income. Higher-income households may have greater access to newer vehicle technologies due to affordability while lower-income households may more heavily rely on conventional gasoline powered vehicles. To investigate this question, variables relating to household income and vehicle fuel type were extracted from the NHTS dataset. Fuel types were grouped into a broader category (Alternative and Conventional) to simplify comparison. This approach allows for a clear understanding of how AFV adoption varies across income groups.

Research Question 2

How does travel behavior, measured by day distribution, vary across household income groups in the Midwest?

Travel behavior can vary significantly based on socioeconomic conditions, including how lifestyle differences, access to transportation, and work schedules affect traffic loads. Understanding these patterns can provide meaningful insight into mobility needs for transportation planning. To address this question, this analysis uses the travel_day variable to examine how travel activity is distributed across the days of the week for income groups. Proportional comparisons were used instead of raw data counts to account for differences in group size.

Methodes

The analysis in this study is based on data from the National Household Travel Survey (NHTS) 2022. The NHTS detest provides detailed information on household demographics, vehicle characteristics, travel behavior across the United States. Variables such as household income, vehicle type, fuel type, and travel day, were selected to enable cross level analysis.

All data process was conducted using Python within a Jupyter Notebook environment. The data was first imported and inspected to understand its structure, including variable types and missing values. To ensure consistency in the analysis, all rows containing missing values were removed. The data was then filtered to include only data pertaining to the Midwest region using the census_region variable. Alingin the analysis with the clients' focus. A copy of the filtered dataset was created to avoid potential issues during data manipulation

To support comparative analysis several variables were transformed and simplified. Income groups were classified into categorical ranges. Low ($0-$49,999), Middle ($50,000-$149,999) and High (<$150,000). Fuel types were standardized into Conventional (gasoline and diesel), Alternative (hybrid and electric), and Other. These groupings simplify the analysis for clear trend comparison.

Specifically, this analysis focuses on comparing patterns across income groups using aggregated summaries and visualizations. Four primary comparisons/visualizations were created

1. Fuel Type Counts by Income (Bar Chart)
    a. Counts the total amount of vehicles across income groups and sorts them into AFV and Conventional
2. Alternative fuel Adoption Rate by Income Group (Bar Chart)
    a. Represents the adoption rate of AFV's as a percentage across income groups
3. Travel Day Distribution (Bar Chart)
    a. A proportional representation of travel activity across days of the week for different income groups
4. Vehicle Type Distribution by Income Group (Stacked Bar Chart)
    a. A visualization to examine vehicle ownership composition across income categories.

The analysis was conducted using multiple Python libraries:

- Pandas for data management
- Matplotlib for data visualization
- Plotly for increase visualization customization
- Seabron for graphical analysis
- NumPy for mathematic operators

All data cleaning, transformation, aggregations etc. were performed in a structured Jupyter Notebook. The workflow is fully reproducible with clearly documented code and steps allowing this analysis to be replicated or expanded in the future.

Results

With the data being prepared and key variables constructed the analysis proceeds by examining patterns across income groups using a series of visualizations and aggregated comparisons. The following presents the results of this analysis by highlighting the relationships between household income, vehicle characteristics, and travel behavior.

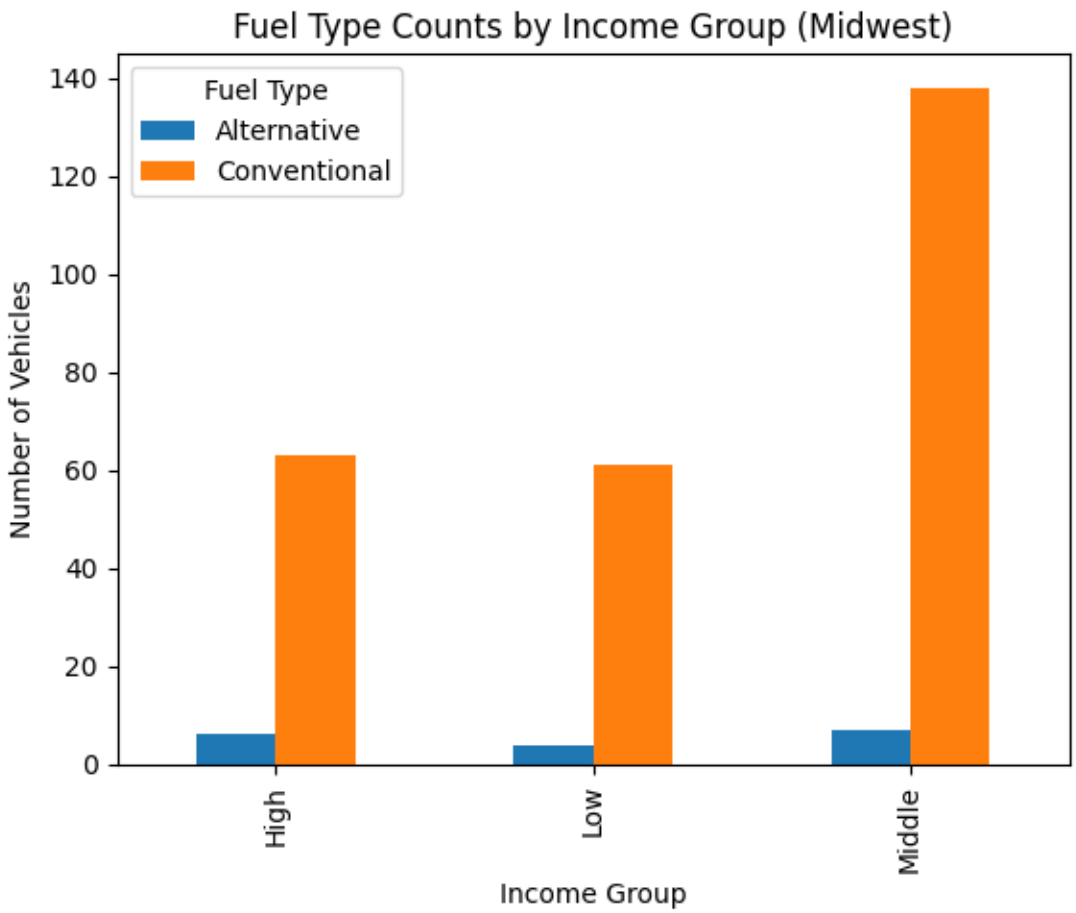

Figure 1

Figure 1 presents the distribution of vehicle fuel types across household income groups in the Midwest. The results show a clear dominance of conventional fuel vehicles over AFV across all income levels indicating that the transition to alternative fuel technologies remains limited. The Middle-income group exhibits the highest overall number of vehicles suggesting that middle income households represent a large share of vehicle ownership withing the dataset. When adjusting focus to alternative fuel vehicles, the Middle-income group has the highest raw count, followed closely by the High-income group, while the Low-income group has the fewest. However raw counts alone do not fully capture adoption patterns

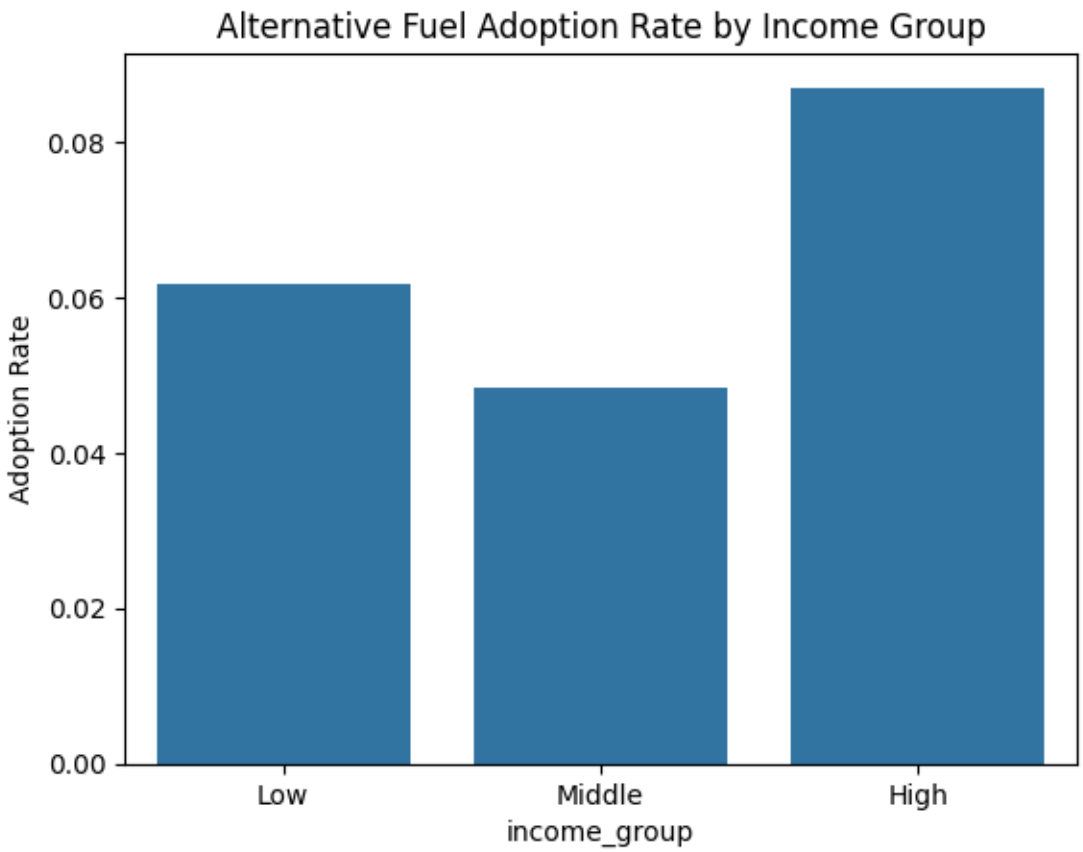

Figure 2

Figure 2 represents a proportional visualization of total alternative fuel vehicles within each income group. High-Income households demonstrate a significantly higher adoption rate compared to Middle income households. These patterns support the expectation that higher income households are more likely to adopt newer technologies due to the greater finical capacity. In contrast Low-income households have a significantly reduced adoption rate reflecting cost barrier and reliance on more affordable options. The most significant findings are the scale of disparity between conventional and AFVs. Even among high income households where adoption rater are highest AFVs represent only a small portion of total vehicles. While income plays a role in adoption, broader structural factors such as electrical infrastructure may also limit the widespread transition to alternative fuel technologies in the Midwest

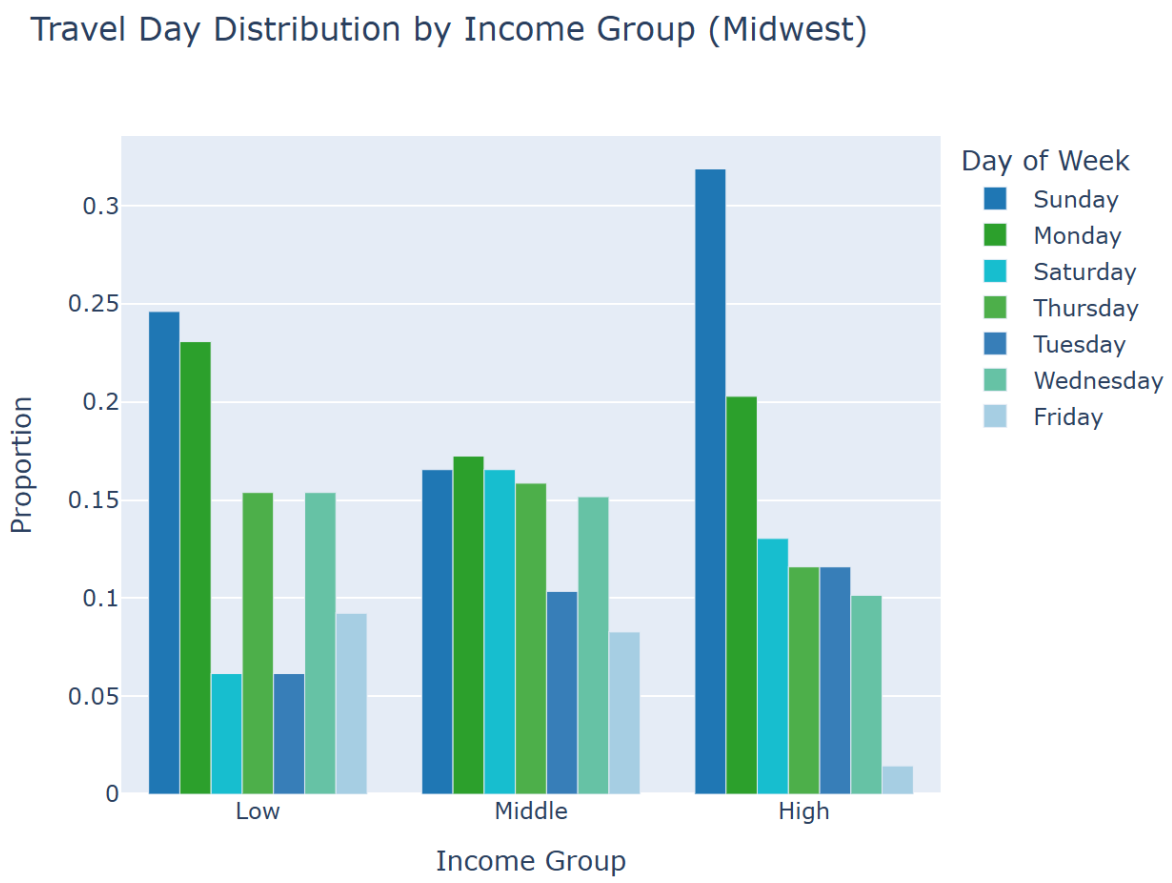

Figure 3

Figure 3 illustrates how travel activity is distributed across the days of the week for different household incomes across the Midwest, The results reveal noticeable differences in both consistency and timing of travel behavior across income levels. The most prominent outlier is the spike in Sunday travel among high income households. This suggests that high income individuals may concentrate a larger share of their travel on the weekends likely for leisure, social events, or errands. Middle income households exhibit a more balanced and consistent distribution of travel across the week with stable counts thought the week, indicating routine patterns that reflect standard work schedules, commuting behavior and moderate weekend activity. Low income households display greater variability in travel. Higher travel activity is observed on Sunday and Monday. This indicates that travel for this group is more closely tied to specific needs or irregular scheduling such as shift based work or constrained access to transportation. Across all income groups, Friday is noticeably the lowest traveled day. This trend may reflect evolving work patterns such as flexible schedules or remote work arrangements which

are frequently associated with higher income occupations. Overall these findings show that income level influences travel volume, and when individuals travel throughout the week, providing meaningful insight to transportation planning to better design future infrastructure to met the fluctuating needs of the community

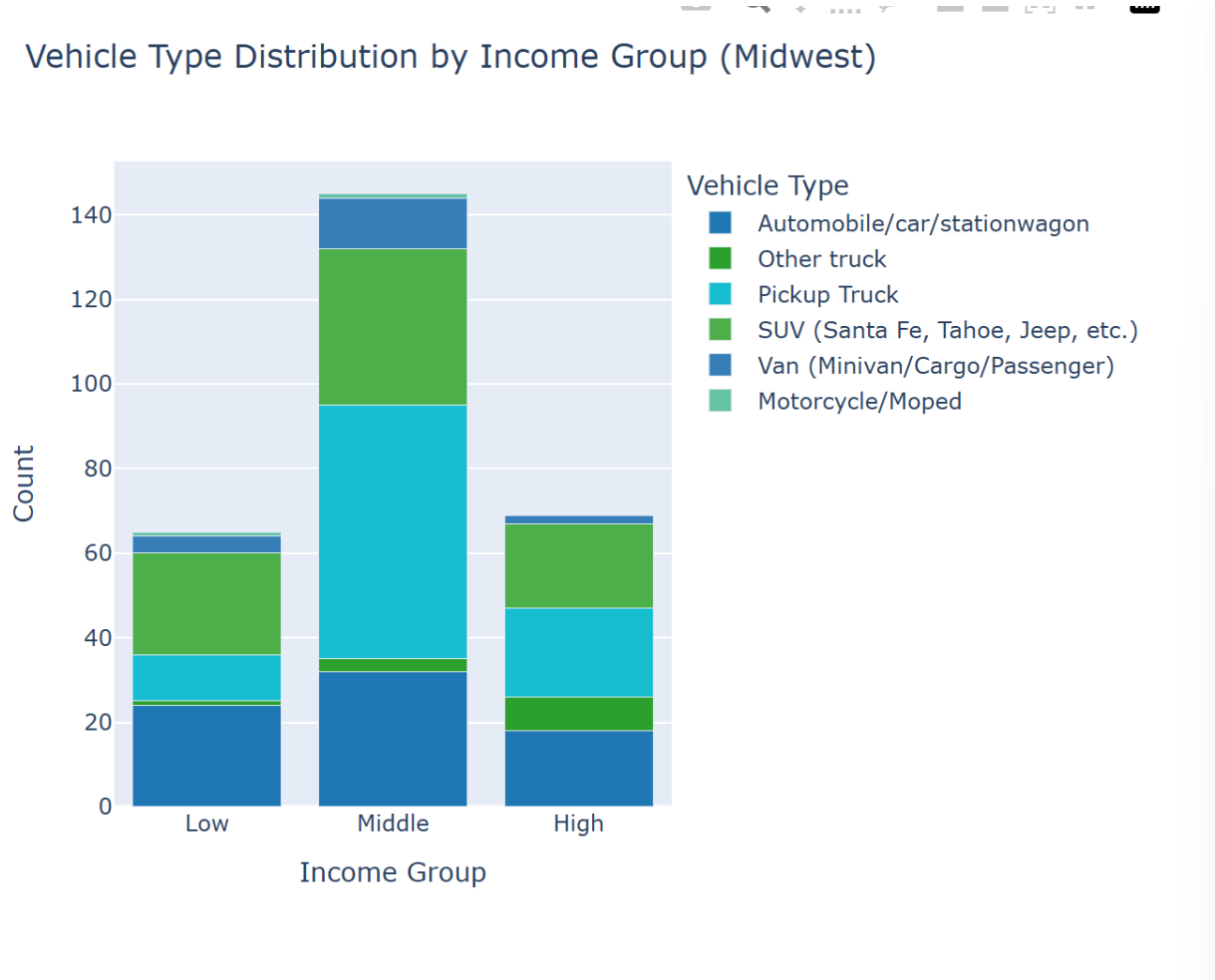

Figure 4

Figure 4 visualizes the distribution of vehicle types across household income groups using a stacked bar chart. The results highlight the differences in vehicle ownership across economic considerations. Notably the middle-income group has a dominance of pickup trucks. This category represents the single largest segment across all income levels. This trend likely reflects the practical utility pickups provided in the Midwest where weather conditions, and occupational needs demand a versatile vehicle. In contrast, high income households show a large make up of SUV's which offer a combination of comfort space and performance reflecting high income lifestyle. Low income households exhibit a higher proportion reliance on standard automobiles such as sedans and station

wagons. When considering this data alongside that of previous figures these result provide important context for the study's questions. The strong presence of pickups among midwestern households explains a potential reasoning for the lower adoption rate of AFVs, as electric and hybrid options for this vehicle type are limited. These findings play an important role in shaping both alternative fuel adoption and travel behavior, Differences in vehicle composition across income groups represent finical capacity but also functional needs

Bais and Limitations

This analysis provides useful insight into travel behavior and vehicle characteristics across income groups, however several limitations and potential sources of bias should be acknowledged. First, the classification of household income in three broad categories simplifies a more complex distribution. This grouping may mask important differences within these ranges and influence how trends appear across categories. The categorization of fuel types also requires assumptions based on the text labels within the dataset. While efforts to standardize these results were made, inconsistences may lead to misclassification of certain vehicle types. Further this data was filtered to only include observations with complete data which resulted in the removal of rows containing missing values. While this technique improves consistency it also introduces unintentional bias.

Conclusion

This analysis explored how household income influences vehicle ownership patterns, and travel behavior in the Midwest using data for the National Household Travel Survey. Across all areas, clear relationships emerged between socioeconomic status and transportation choices. The results show how conventional fuel vehicles continue to dominate across all income groups, with alternative fuel vehicles remaining relatively low across all incomes. Higher incomes demonstrate a greater proportion of alternative fuel vehicles but the overall transition to these technologies is still limited. Additionally, differences in vehicle type such as the prevalence of pickup trucks among middle income households highlight the role of functionally and regional demands which shape vehicle ownership composition. Trevel behavior analysis further revealed that income influences more than what populations drive. It has a direct effect on travel patterns. Higher income households can be observed to have concentrated travel on weekends while middle income households exhibit more consistent weekly patterns, and low-income households show a greater variability. These differences suggest that work structure, flexibility, and access to resources all play a role in mobility patterns. The differences in travel behavior across income groups reveals the importance of designing transportation systems that

accommodate a wide range of mobility needs. Flexible transit options, equitable access to infrastructure, and consideration of non-traditional travel patterns are all essential for planning efficient, and useful transportation systems. By incorporating these insights into planning and policy decisions, transportation engineers can better design systems that are responsive to the diverse needs of the population.

Works Cited

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